# Vivek Singh Roll NO – 2501940023 Course Code – ETMMDL274

**Deep Learning Capstone Project Report**

## 1. Introduction

This capstone project focuses on implementing and analyzing various deep learning architectures and techniques learned during the course Deep Learning.

The project includes:

- Feedforward Neural Network (FNN) for classification
- Convolutional Neural Network (CNN) for image classification
- Long Short-Term Memory (LSTM) for sequence modeling

These experiments demonstrate the practical applications of deep learning in different domains.

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## 2. Methodology

### Experiment 1: Feedforward Neural Network

**Dataset Used:** UCI Adult Income Dataset

**Preprocessing Steps:**

- Missing values handled
- Categorical values encoded
- Features normalized
- Train-test split applied

**Model Architecture:**

- Input Layer
- Dense Layer (128 neurons, ReLU)
- Dropout Layer
- Dense Layer (64 neurons, ReLU)
- Output Layer (Sigmoid)

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### Experiment 2: CNN for Image Classification

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**Dataset Used:** Fashion-MNIST

**Preprocessing:**

- Normalized pixel values
- Reshaped images

**Model Architecture:**

- Conv2D + MaxPooling  Conv2D + MaxPooling
- Flatten
- Dense + Dropout
- Output Layer (Softmax)

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**Experiment 3: LSTM for Sequence Modeling**

**Dataset Used:** Stock Price Dataset

**Preprocessing:**

- Closing price selected
- Normalization
- Sequence windowing

**Model Architecture:**

- LSTM Layer
- LSTM Layer
- Dense Output Layer

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**3. Implementation Details**

**Framework Used:** TensorFlow / Keras

| Hyperparameter | Value |
|---|---|
| Optimizer | Adam |
| Epochs | 10–20 |
| Batch Size | 32 |

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**Hyperparameter Value**

| | |
|---|---|
| Loss Function | CrossEntropy / MSE |

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**4. Results**

**Experiment 1:**

- Accuracy achieved: 85% approx

**Experiment 2:**

- Accuracy achieved: 88% approx
- Precision, Recall, F1-score calculated
- Confusion Matrix generated

**Experiment 3:**

- RMSE calculated for prediction performance
- Actual vs Predicted graph generated

Graphs included:

- Training vs Validation Accuracy
- Training vs Validation Loss

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**5. Discussion**

**Observations:**

- FNN works well for structured/tabular data
- CNN performs effectively on image classification
- LSTM captures temporal dependencies in sequence data

**Challenges Faced:**

- Overfitting in CNN
- Data preprocessing complexity
- Hyperparameter tuning

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**Comparison:**

**Model Best For**

| | |
|---|---|
| FNN | Tabular Data |
| CNN | Images |
| LSTM | Time-Series / Sequential Data |

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**6. Conclusion**

This project provided hands-on experience with multiple deep learning architectures.

Key learnings:

- Designing neural networks
- Model evaluation techniques
- Handling overfitting
- Real-world applicability of deep learning

Deep learning models can be effectively applied in finance, healthcare, image recognition, and forecasting systems.

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**7. References**


- UCI Machine Learning Repository
- TensorFlow Documentation
- Keras Documentation
- Research papers and online tutorials